

Developer Mock Test Results

Student: Shiva kasula

Name:	Shiva kasula	Student ID:	112
Date:	2026-03-05 12:01:19	Performance:	Needs Improvement
Overall Score:	0.0/25 (0.0%)	Test ID:	d24ea41b-ecec-49cc...
Warnings:	0/3	Status:	Completed

Section-wise Performance

Section	Score	Percentage	Status
Aptitude	0/10	0.0%	Needs Work
MCQ/Theory	0/10	0.0%	Needs Work
Coding	0.0/5	0.0%	Needs Work

Detailed Question Review

APTITUDE SECTION (0/10 - 0.0%)

Q1. If 5 men can build a wall in 10 days, how many men are needed to build the same wall in 5 days?

Your Answer: No answer (Skipped)

Correct Answer: 10

To find the number of men needed to build the wall in 5 days, we can use the formula: $\text{men} \times \text{days} = \text{constant}$. Given that 5 men can build the wall in 10 days, we have $5 \times 10 = 50$ man-days. To build the same wall in 5 days, we need $50 \text{ man-days} / 5 \text{ days} = 10$ men. The correct answer is indeed 10, as calculated. If the user got it wrong, their likely mistake was not using the correct inverse proportionality between the number of men and the number of days.

Q2. If 3 men can build a wall in 8 days, how many men are needed to build the same wall in 4 days?

Your Answer: No answer (Skipped)

Correct Answer: 6

To determine the number of men needed to build the wall in 4 days, we can use the formula: $\text{Men} \times \text{Days} = \text{Constant}$. Given that 3 men can build the wall in 8 days, we have $3 \times 8 = 24$ man-days. To build the same wall in 4 days, we need to find the number of men that satisfies the equation: $\text{Men} \times 4 = 24$. The key calculation step is: $\text{Men} = 24 / 4 = 6$. This calculation shows that 6 men are needed to build the wall in 4 days, making option B the correct answer.

Q3. If 18 workers can build a house in 12 days, how many workers will be needed to build the same house in 10 days?

Your Answer: No answer (Skipped)

Correct Answer: 22 workers

To find the number of workers needed to build the house in 10 days, we can use the formula: $\text{workers} \times \text{days} = \text{constant}$. Given that 18 workers can build the house in 12 days, we have $18 \times 12 = 216$ worker-days. To build the same house in 10

days, we need $216 \text{ worker-days} / 10 \text{ days} = 21.6 \text{ workers}$, which we round up to the nearest whole number since we can't have a fraction of a worker, suggesting a calculation error in the provided correct answer. The correct calculation is $216 \text{ worker-days} / 10 \text{ days} = 21.6 \text{ workers}$, which is closest to option D) 21 workers when rounded to the nearest whole number, not 22 workers as stated. The likely mistake in the provided correct answer is a calculation error.

■ **Q4. The average of 6 numbers is 18. If one of the numbers is 20, what is the sum of the remaining 5 numbers?**

Your Answer: No answer (Skipped)

Correct Answer: 100

■ To find the sum of the remaining 5 numbers, we first calculate the total sum of the 6 numbers by multiplying the average by the total count: $18 * 6 = 108$. Since one of the numbers is 20, we subtract this from the total sum to find the sum of the remaining 5 numbers: $108 - 20 = 88$. However, the correct answer is stated as 100, but based on the calculation, the correct sum should be 88, not 100.

■ **Q5. If it is true that all students are learners and some learners are teachers, what can be concluded?**

Your Answer: No answer (Skipped)

Correct Answer: Some learners are students

■ The correct answer is indeed "Some learners are students" because we know that all students are learners, which implies that the set of students is a subset of the set of learners. This means that at least some learners are students. The user's answer, "Some students are teachers", is incorrect because while it may be true, it doesn't directly follow from the given statements. The given statements only guarantee that some learners are teachers and all students are learners, but they don't provide enough information to conclude that some students are teachers.

■ **Q6. The average of 3 numbers is 15. What is the sum of the 3 numbers?**

Your Answer: No answer (Skipped)

Correct Answer: 45

■ To find the sum of the 3 numbers, we need to multiply the average by the total number of values, which is 3. The key calculation step is: $\text{average} * \text{number of values} = \text{sum}$, so $15 * 3 = 45$. Therefore, the correct answer is indeed 45, and the user fortunately chose the correct option.

■ **Q7. The average of 4 numbers is 25. If one of the numbers is 30, what is the sum of the remaining 3 numbers?**

Your Answer: No answer (Skipped)

Correct Answer: 70

■ The correct answer is 70. To find the sum of the remaining 3 numbers, we first calculate the total sum of the 4 numbers: $\text{average} * \text{number of numbers} = 25 * 4 = 100$. Then, we subtract the given number (30) from the total sum: $100 - 30 = 70$, which is the sum of the remaining 3 numbers.

■ **Q8. If a number is increased by 25% and then decreased by 25%, what is the overall percentage change?**

Your Answer: No answer (Skipped)

Correct Answer: -6.25%

■ To find the overall percentage change, let's consider a number, say 100. When increased by 25%, it becomes $100 + (100 * 0.25) = 125$. Then, when decreased by 25%, it becomes $125 - (125 * 0.25) = 93.75$. The key calculation step is: $(100 * 0.25) = 25$ for the increase, and $(125 * 0.25) = 31.25$ for the decrease. The overall change is $93.75 - 100 = -6.25$, which is a decrease of 6.25%. If the user got it wrong, their likely mistake was assuming that increasing and then decreasing by the same percentage would result in the original number, which is not the case due to the different bases for the percentage calculations.

■ **Q9. If 12 workers can build a house in 8 days, how many workers will be needed to build the same house in 6 days?**

Your Answer: No answer (Skipped)

Correct Answer: 16 workers

■ To find the number of workers needed, we can use the formula: $\text{workers} * \text{days} = \text{constant}$. Given that 12 workers can build a house in 8 days, we have $12 * 8 = 96 \text{ worker-days}$. To build the same house in 6 days, we need to find the number of workers that gives us 96 worker-days, so $\text{workers} * 6 = 96$. The key calculation step is: $\text{workers} = 96 / 6 = 16$. This calculation confirms that the correct answer is indeed 16 workers, as it would take the same amount of total work (96 worker-days) to build the house in 6 days with more workers.

■ **Q10. If 6 workers can build a house in 10 days, how many workers will be needed to build the same house in 8 days?**

Your Answer: No answer (Skipped)

Correct Answer: 8 workers

■ To find the number of workers needed to build the house in 8 days, we can use the formula: (number of workers) * (number of days) = constant. Given that 6 workers can build the house in 10 days, we have $6 * 10 = 60$ worker-days. To build the same house in 8 days, we need $60 \text{ worker-days} / 8 \text{ days} = 7.5$ workers. Since the question likely assumes whole workers and the provided options do not include fractions, the closest whole number is 7.5, which would be rounded, but given the context of the question and the provided options, it seems the intention might have been to find the nearest whole number that can accomplish the task without going under, thus possibly leading to considering 8 workers as a practical answer due to the nature of not being able to have half a worker. However, based on the calculation, the exact number of workers needed would be 7

■ MCQ SECTION (0/10 - 0.0%)

■ **Q1. What is a deadlock in Java?**

Your Answer: No answer (Skipped)

Correct Answer: A situation where two or more threads are blocked indefinitely

■ The correct answer is A) A situation where two or more threads are blocked indefinitely. This is because a deadlock in Java occurs when two or more threads are unable to proceed with their execution due to a circular dependency on resources, resulting in a permanent blockage. The user likely got it right, as this definition accurately describes the concept of a deadlock, where multiple threads are stuck waiting for each other to release resources, leading to a permanent standoff.

■ **Q2. What is optimization in Java?**

Your Answer: No answer (Skipped)

Correct Answer: The process of improving the performance of a program

■ The correct answer is A) The process of improving the performance of a program. This is because optimization in Java refers to the process of modifying a program to make it more efficient, reducing its execution time, and improving its overall performance. The user's likely mistake, if they chose a different option, would be confusing optimization with other aspects of program development, such as adding new features (functionality) or enhancing security measures.

■ **Q3. What is an example of a data type in Java?**

Your Answer: No answer (Skipped)

Correct Answer: Int

■ The correct answer is C) Int because it is a fundamental data type in Java, used to represent whole numbers. Int is a primitive data type that allows variables to store integer values, making it a basic building block of Java programming. The user's likely mistake, if they chose a different option, is confusing control structures (if-else, for loop) or program components (method) with a data type, which is a category of data that a variable can hold.

■ **Q4. What are the operators used in Java?**

Your Answer: No answer (Skipped)

Correct Answer: All of the above

■ The correct answer is "All of the above" because Java utilizes a variety of operators to perform different operations. These include arithmetic operators for mathematical calculations, comparison operators for evaluating conditions, and logical operators for making decisions based on multiple conditions. The user likely got it right, as Java indeed supports all these types of operators, making "All of the above" the most comprehensive and accurate choice.

■ **Q5. What is a RuntimeException in Java?**

Your Answer:

No answer (Skipped)

Correct Answer:

A type of exception that is thrown at runtime

■ The correct answer is A) A type of exception that is thrown at runtime. This is because a RuntimeException in Java is a subclass of the Exception class that is thrown at runtime, meaning it occurs during the execution of the program, rather than at compile-time. The user likely got it right, as this is a fundamental concept in Java, but if they got it wrong, their mistake was probably confusing runtime exceptions with compile-time errors or exceptions.

■ Q6. What is an example of a simple Java program?

Your Answer: No answer (Skipped)

Correct Answer: All of the above

■ The correct answer is "All of the above" because a simple Java program can indeed encompass a wide range of applications, including calculating student grades, managing simple data, and printing a greeting message. These examples represent basic functionalities that can be achieved with minimal code and complexity, making them ideal for introductory Java programming. The user's answer is not provided, but if they chose a single option, their likely mistake would be underestimating the breadth of what constitutes a simple Java program.

■ Q7. What is an example of a control structure in Java?

Your Answer: No answer (Skipped)

Correct Answer: If-else statement

■ The correct answer is C) If-else statement because it is an example of a control structure in Java, which determines the flow of a program's execution based on conditions or decisions. Control structures, such as if-else statements, allow a program to execute different blocks of code under different circumstances. The user's likely mistake, if they chose a different option, would be confusing a control structure with a fundamental programming concept, such as variables, data types, or classes, which serve different purposes in Java.

■ Q8. What is encapsulation in Java?

Your Answer: No answer (Skipped)

Correct Answer: The concept of hiding internal details and showing only necessary information

■ The correct answer is A) The concept of hiding internal details and showing only necessary information. This is because encapsulation in Java is a fundamental principle of object-oriented programming that involves bundling data and methods that operate on that data within a single unit, while hiding the internal implementation details from the outside world. The user's likely mistake, if they chose a different option, would be misunderstanding the concept of encapsulation, which is essentially about controlling access to an object's internal state by exposing only the necessary information through public methods.

■ Q9. What is a livelock in Java?

Your Answer: No answer (Skipped)

Correct Answer: A situation where two or more threads are unable to access a shared resource

■ The correct answer is A) A situation where two or more threads are unable to access a shared resource. This is because a livelock in Java occurs when two or more threads are unable to access a shared resource due to constant retries and failures, often caused by poor synchronization or resource allocation. The user likely got it right, as this definition accurately describes a livelock scenario, but if they got it wrong, their mistake was probably due to confusing it with a deadlock, which is a related but distinct concept where threads are blocked and unable to proceed.

■ Q10. What are the types of errors in Java?

Your Answer: No answer (Skipped)

Correct Answer: All of the above

■ The correct answer is "All of the above" because Java has three main types of errors: syntax errors, which occur when the code does not follow the language's rules; logical errors, which occur when the code does not produce the desired result due to a flaw in the program's logic; and runtime exceptions, which occur when an error happens during the execution of the code. These three types of errors can all occur in Java, making "All of the above" the correct answer. The user likely got it right, as the question is straightforward and the options are comprehensive.

■ CODING SECTION (0.0/5 - 0.0%)

■ Q1. Write a Java program that reads an array of integers from stdin and rotates it by a specified number of positions. ****Input Format:**** - Line 1: The nu...

Your Answer:

No answer (Skipped)

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        int n = scanner.nextInt();
        int[] array = new int[n];

        for (int i = 0; i < n; i++) {
            array[i] = scanner.nextInt();
        }

        int k = scanner.nextInt();
        k = k % n; // handle cases where k > n

        int[] rotatedArray = new int[n];

        // copy last k elements to the beginning of the rotated array
        System.arraycopy(array, n - k, rotatedArray, 0, k);

        // copy the remaining elements to the rotated array
        System.arraycopy(array, 0, rotatedArray, k, n - k);

        // print the rotated array
        for (int i = 0; i < n; i++) {
            System.out.print(rotatedArray[i] + " ");
        }

        scanner.close();
    }
}
```

■ There are no test case results provided to analyze the issue with the student's code. No feedback can be given without the actual test case results. Please provide the test case results to determine the cause of the failure.

■ Q2. Write a Java program that reads an integer from stdin and checks if it's within a valid range (1-100).
Input Format: - Line 1: An integer **Outp...

Your Answer:

No answer (Skipped)

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int number = scanner.nextInt();
        scanner.close();

        if (number >= 1 && number <= 100) {
            System.out.println("Valid");
        } else {
            System.out.println("Invalid");
        }
    }
}
```

■ There is no test case failure provided to analyze. The student's code is also not submitted, so it's impossible to determine what went wrong. Without the actual test case results or the student's code, it's not possible to provide accurate feedback.

■ **Q3. Write a program that reads a string and an integer, and prints the result of dividing the integer by the length of the string. Handle any potential er...**

Your Answer:

No answer (Skipped)

Correct Answer:

```
def divide_integer_by_string_length():
    string = input()
    try:
        integer = int(input())
        if len(string) == 0:
            print("Error: Division by zero is not allowed")
        else:
            print(integer / len(string))
    except ValueError:
        print("Error: Invalid input")

divide_integer_by_string_length()
```

■ There are no test case results provided to analyze the issue. Since no test case results or student code is submitted, it's impossible to determine what went wrong. No feedback can be given based on the provided information.

■ **Q4. Write a program that reads a dictionary of word frequencies and prints the word with the highest frequency. **Input Format:** - Line 1: Number of wor...**

Your Answer:

No answer (Skipped)

Correct Answer:

```
n = int(input())
word_freq = {}
for _ in range(n):
    word, freq = input().split()
    word_freq[word] = int(freq)
print(max(word_freq, key=word_freq.get))
```

■ There is no test case failure information provided to determine what went wrong. The student's code and test case results are not available for analysis. Without this information, it's impossible to provide accurate feedback on the code's performance.

■ **Q5. Write a Java program that defines a Shape interface with methods to calculate area and perimeter, and a Circle class that implements Shape. **Input F...**

Your Answer:

No answer (Skipped)

Correct Answer:

```
import java.util.Scanner;

interface Shape {
    double calculateArea();
    double calculatePerimeter();
}

class Circle implements Shape {
    private double radius;

    public Circle(double radius) {
        this.radius = radius;
    }
}
```

```

    }

    @Override
    public double calculateArea() {
        return Math.PI * Math.pow(radius, 2);
    }

    @Override
    public double calculatePerimeter() {
        return 2 * Math.PI * radius;
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int n = scanner.nextInt();
        scanner.nextLine(); // Consume newline left-over

        for (int i = 0; i < n; i++) {
            String shapeType = scanner.next();
            double radius = scanner.nextDouble();
            scanner.nextLine(); // Consume newline left-over

            if (shapeType.equals("Circle")) {
                Circle circle = new Circle(radius);
                System.out.println(String.format("%.1f %.1f", circle.calculateArea(),
circle.calculatePerimeter()));
            } else if (shapeType.equals("Shape")) {
                // Assuming Shape is a rectangle with side length equal to radius
                double area = Math.pow(radius, 2);
                double perimeter = 4 * radius;
                System.out.println(String.format("%.1f %.1f", area, perimeter));
            }
        }

        scanner.close();
    }
}

```

■ There is no test case failure provided, so it's impossible to determine what went wrong. No code was submitted, and no test case results are available for analysis. Without this information, I cannot provide accurate feedback on the student's code.

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
- **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.
- **Coding:** Practice more coding problems on platforms like LeetCode or HackerRank.